AI-BASED SUMMARIZATION AND CLAUSE CLASSIFICATION OF INDIAN LEGAL DOCUMENTS USING NLP AND TRANSFORMER MODELS

Mini-Project Synopsis

Submitted by

Vedant jitani– 240962492
Roll number-41

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Under the guidance of
Dr Rani Ooman Panicker, Professor
Dr H Satyajeet Sharma

School of Computer Engineering
Manipal Institute of Technology
Manipal Academy of Higher Education, Manipal

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1. Title of the Problem

AI-Based Summarization and Clause Classification of Indian Legal Documents Using NLP and Transformer Models.

2. Introduction

India's legal system faces a severe backlog, with millions of pending cases, heavy worded judgments and contracts that are time-consuming for professionals to review manually. Natural Language Processing (NLP) and transformer-based language models such as BERT have shown strong potential for automating legal text analysis,However, Indian legal documents pose unique challenges: they are often unstructured, and stylistically distinct from Western case law. Prior work on domain-specific summarization models and legal transformer variants has demonstrated improved performance over generic NLP when adapted to Indian legal text . This project proposes an NLP-based system that automatically summarizes Indian court judgments and classifies clauses into legal categories, using transformer fine-tuned on Indian legal code. By reducing the manual effort required to review lengthy legal documents, the system aims to support lawyers, legal researchers, and self-represented litigants in navigating India's legal system more efficiently.

3. Keywords

Legal NLP; Document Summarization; Clause Classification; Transformer Models; LegalBERT; Indian Judiciary.

4. Objectives

- To collect and pre-process Indian legal documents while handling noisy OCR text, inconsistent formatting, and document-length variation.
- To design and fine-tune transformer-based models for automatic summarization of judgments and classification of clauses.
- To evaluate model performance using standard metrics — ROUGE for summarization and F1-score/accuracy for classification.

5. Methodology

- Data Acquisition: Collect Indian court judgments from public repositories such as IndianKanoon and the e-Courts National Judicial Data Grid, and contract clauses from open datasets such as CUAD, for classification pretraining.
- Data Pre-processing: Clean and normalise legal text and construct labelled datasets for clause categories.

- Feature Engineering: Generate contextual embeddings using pre-trained legal transformer models (LegalBERT/InLegalBERT) as input representations.
- Model Development: Fine-tune transformer-based summarization models (Legal-Pegasus, BART, Longformer) for judgment summarization, and classifiers for contract clause classification, using Python, Hugging Face Transformers, and PyTorch.
- Model Evaluation: Compare fine-tuned transformer models against classical ML baselines (TF-IDF + SVM/Logistic Regression) using ROUGE, F1-score, and accuracy on held-out Indian legal test data.
- Analysis and Reporting: Analyse results, identify the best-performing configuration, and document findings in the final project report.

6. Data Source

Primary data will be sourced from publicly available Indian legal judgment repositories such as IndianKanoon and the e-Courts National Judicial Data, which provide free-text access to Indian court judgments. For contract clause classification, the project will additionally use open benchmark datasets such as the Contract Understanding Atticus Dataset (CUAD) for pretraining and cross-domain evaluation, supplemented by publicly available Indian legal-document summarization datasets referenced in prior work .

7. Plan of Execution

Phase	Activity
Phase 1	finalisation of datasets, clause taxonomy, and models
Phase 2	Data acquisition (IndianKanoon, e-Courts, CUAD) and data pre-processing
Phase 3	data analysis and dataset labelling
Phase 4	Implementation of baseline models (TF-IDF + SVM / Logistic Regression)
Phase 5	Fine-tuning of transformer models (LegalBERT, Legal-Pegasus/BART)
Phase 6	Comparative evaluation, hyper-parameter tuning
Phase 7	Report writing, documentation, and final presentation

8. References

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